PCOS Prediction

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Introduction

Polycystic Ovary Syndrome (PCOS) is a hormonal disorder that affects some individuals of reproductive age. It is caused by having too much of a hormone called androgen in the body. Early detection is important because symptoms can overlap, making diagnosis difficult. In this project, I built machine learning models to predict PCOS using features such as Body Mass Index (BMI), testosterone level, age, and antral follicle count.

Abstract

The goal of this project was to compare two classification models, Naive Bayes and Decision Tree, for predicting PCOS. The dataset was explored using histograms, bar graphs, and box plots. Before training, the Menstrual_Irregularity variable was removed because it caused a zero variance error. The data was split into 70% training and 30% testing before both models were trained and evaluated.

Materials

The dataset was obtained from Kaggle and contains 1,000 records. The analysis was completed in R using packages including rsample, rpart, rpart.plot, naivebayes, ggplot2, caret, and dplyr. A random seed was used to make the results reproducible.

Methodology

- Loaded the PCOS dataset.
- Converted categorical variables to factors.
- Removed Menstrual_Irregularity because of the zero variance error.
- Performed exploratory data analysis using histograms, bar graphs, and box plots.
- Split the data into 70% training and 30% testing.
- Trained Naive Bayes and Decision Tree models.
- Compared the performance of both models.

Results

The Decision Tree model achieved an accuracy of 89.33%, while the Naive Bayes model achieved 82.33%. The Decision Tree gave better results because it was able to learn relationships between the variables instead of treating them independently.

Discussion

BMI, testosterone level, and antral follicle count were the strongest predictors of PCOS. The Naive Bayes model struggled because of the class imbalance and because many of the features were related to each other. The Decision Tree gave more balanced and accurate predictions.

Conclusion

This project compared two machine learning models, Naive Bayes and Decision Tree, to predict PCOS using patient data. Both models were able to classify patients, but the Decision Tree performed better with an accuracy of 89.33% compared to 82.33% for Naive Bayes.

From this project, I learned that choosing the right model is important because different models perform differently on the same dataset. I also learned how data preprocessing, exploratory data analysis, and model evaluation affect the final results. Overall, the Decision Tree gave better results on this dataset and predicted PCOS more accurately than the Naive Bayes model.

Future Improvements

If I had more time, I would like to try more machine learning models and compare their performance with the models used in this project. I would also like to use a larger dataset and include more patient information to see if it improves the prediction results. Another improvement would be to build a simple application where users could enter patient information and receive a prediction from the trained model.

References

1. Kaggle PCOS Diagnosis Dataset
https://www.kaggle.com/datasets/samikshadalvi/pcos-diagnosis-dataset

2. World Health Organization (WHO) – Polycystic Ovary Syndrome
https://www.who.int/news-room/fact-sheets/detail/polycystic-ovary-syndrome